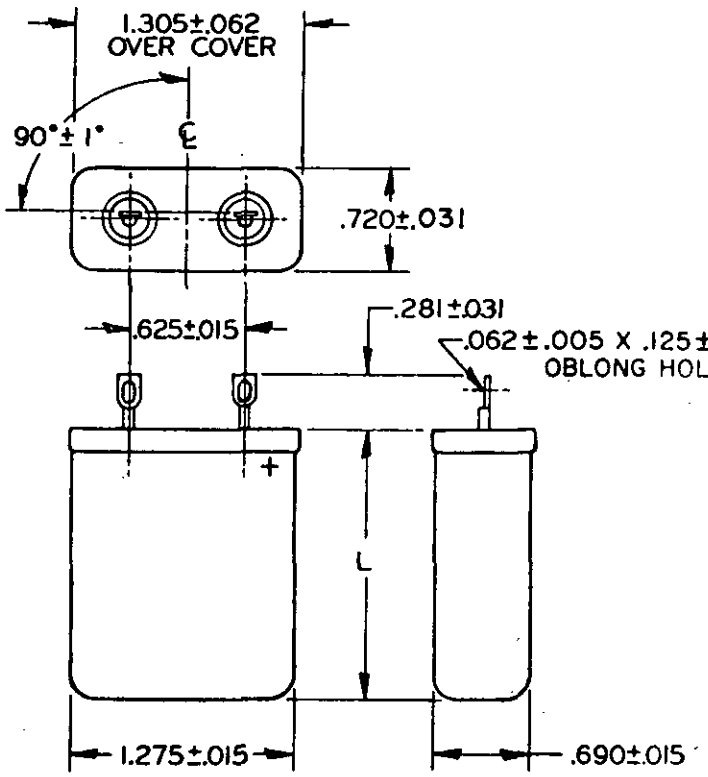
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM ANODE AND CATHODE, (POLARIZED, SINTERED SLUG) - 85°C (VOLTAGE DERATED TO 125°C), STYLE CL56

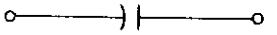
This specification is approved for use by Engineering and Technical Support Division, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-3965.



Case size	Dimensions	
	L ±.031 (.79)	
	Inch	MM
A1	1.062	26.97
A2	1.375	34.93
A3	1.625	41.28
A4	2.000	50.80
A5	2.500	63.50

INCHES	MM	INCHES	MM
.015	.38	.625	15.88
.031	.79	.690	17.53
.062	1.57	.720	18.29
.125	3.18	1.275	32.39
.281	7.14	1.305	33.15



CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Style CL56 capacitors.

FSC 5910

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TABLE I. Capacitor characteristics.

Type designation	Voltage			Capacitance 1/	DC leakage (max)		AC ripple (85°C)	Impedance (max)	Case size
	Rated (85°C)	Derated (125°C)	Surge (85°C)		25°C	85°C & 125°C			
	volts,dc	volts,dc	volts,dc	µF	µA	µA	A	ohms	
CL56BE961MPG	15	10	17.2	960	7	58	3.5	3.10	A1
CL56BE112MPG	15	10	17.2	1100*	16	64	4.0	2.30	A1
CL56BE122MPG	15	10	17.2	1200	9	72	4.0	2.30	A2
CL56BE142MPG	15	10	17.2	1400	11	84	5.0	1.70	A3
CL56BE162MPG	15	10	17.2	1600*	22	88	5.5	1.70	A2
CL56BE202MPG	15	10	17.2	2000*	28	112	7.0	1.35	A3
CL56BE212MPG	15	10	17.2	2100	16	126	7.5	1.30	A4
CL56BE242MPG	15	10	17.2	2400	18	144	8.5	1.15	A5
CL56BE262MPG	15	10	17.2	2600*	36	144	9.0	1.10	A4
CL56BE332MPG	15	10	17.2	3300*	46	184	11.5	0.80	A5
CL56BH521MPG	30	20	34.5	520	8	63	2.5	5.30	A1
CL56BH661MPG	30	20	34.5	660	10	80	3.5	4.20	A2
CL56BH721MPG	30	20	34.5	720*	16	64	4.0	3.40	A1
CL56BH821MPG	30	20	34.5	820	13	99	4.5	2.90	A3
CL56BH102MPG	30	20	34.5	1000*	22	88	5.5	2.80	A2
CL56BH122MPG	30	20	34.5	1200	18	144	6.5	2.30	A4
CL56BH132MPG	30	20	34.5	1300*	28	112	7.0	2.10	A3
CL56BH172MPG	30	20	34.5	1700*	36	144	9.0	1.70	A4
CL56BH222MPG	30	20	34.5	2200*	46	184	11.5	1.30	A5
CL56BJ401MPG	50	30	57.5	400	10	80	3.5	7.20	A1
CL56BJ431MPG	50	30	57.5	430*	10	78	4.0	6.40	A1
CL56BJ501MPG	50	30	57.5	500	13	100	5.0	5.60	A2
CL56BJ551MPG	50	30	57.5	550*	22	88	5.5	4.90	A2
CL56BJ601MPG	50	30	57.5	600	15	120	6.0	4.10	A3
CL56BJ701MPG	50	30	57.5	700*	28	112	7.0	3.80	A3
CL56BJ801MPG	50	30	57.5	800	20	160	8.0	3.10	A4
CL56BJ901MPG	50	30	57.5	900*	36	144	9.0	3.00	A4
CL56BJ102MPG	50	30	57.5	1000	23	180	10.0	2.80	A5
CL56BJ122MPG	50	30	57.5	1200*	46	184	11.5	2.30	A5
CL56BL271MPG	75	50	86.2	270*	9	81	4.0	8.50	A1
CL56BL331MPG	75	50	86.2	330	12	99	5.0	7.00	A2
CL56BL371MPG	75	50	86.2	370*	22	88	5.5	6.70	A2
CL56BL401MPG	75	50	86.2	400	15	119	6.0	5.00	A3
CL56BL471MPG	75	50	86.2	470*	28	112	7.0	5.00	A3
CL56BL601MPG	75	50	86.2	600*	23	180	9.0	3.65	A4
CL56BL661MPG	75	50	86.2	660	25	198	10.0	3.50	A5
CL56BL781MPG	75	50	86.2	780*	46	184	11.5	3.10	A5
CL56BN171MPG	100	65	115.0	170	9	68	3.0	15.00	A1
CL56BN221MPG	100	65	115.0	220*	16	64	4.0	10.90	A1
CL56BN261MPG	100	65	115.0	260	13	104	4.5	8.00	A3
CL56BN301MPG	100	65	115.0	300*	22	88	5.5	7.80	A2
CL56BN351MPG	100	65	115.0	350	18	140	6.5	6.50	A4
CL56BN381MPG	100	65	115.0	380*	28	112	7.0	6.10	A3
CL56BN441MPG	100	65	115.0	440	22	176	8.0	5.80	A5
CL56BN491MPG	100	65	115.0	490*	36	144	9.0	4.80	A4
CL56BN621MPG	100	65	115.0	620*	46	184	11.5	3.70	A5
CL56BP161MPG	125	100	144.0	160*	16	64	4.0	14.80	A1
CL56BP221MPG	125	100	144.0	220*	22	88	5.5	10.90	A2
CL56BP281MPG	125	100	144.0	280*	28	112	7.0	8.40	A3
CL56BP361MPG	125	100	144.0	360*	36	144	9.0	6.70	A4
CL56BP461MPG	125	100	144.0	460*	46	184	11.5	5.00	A5

1/ The asterisk (*) indicates high-capacitance series.

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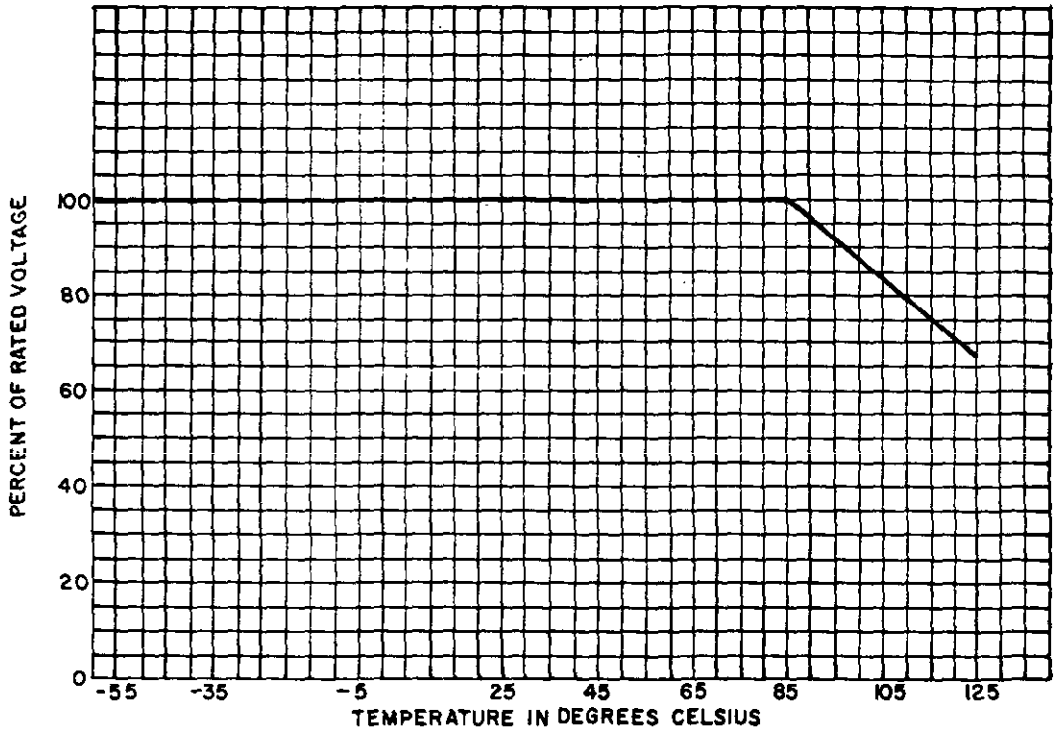


FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:
Dimensions and configuration:
Capacitors - See figure 1.
Brackets - See figure 3.
Case type - Rectangular, uninsulated.
Seal type - Hermetic (symbol G).
Terminals - See figure 1.
Rated temperature - -55°C to +85°C, voltage derated to +125°C (see figure 2).
Rated voltage - See table I.
Capacitance (cap) (nom) - See table I.
Capacitance tolerance - Symbol M (±20%).

DC leakage (DCL) (max) (at +25°C):
See table I.

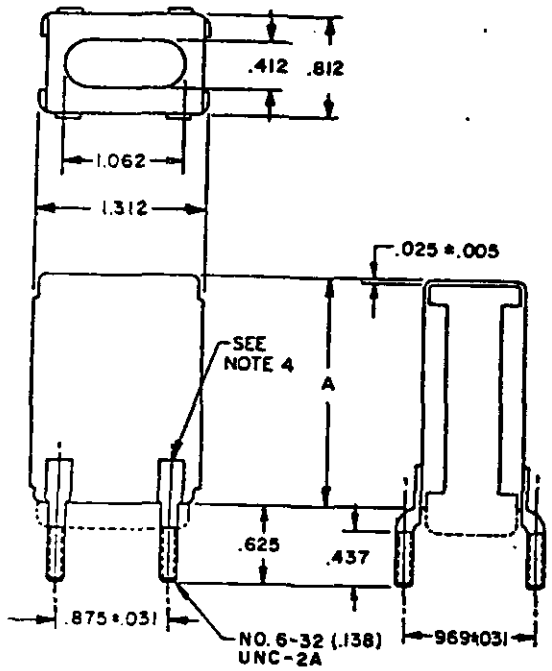
Dissipation factor (DF) (max):
17%(15V); 12%(30V); 10%(50V); 8%(75V); 7%(100V); and 5%(125V).

Seal: In accordance with MIL-C-3965.

Dielectric withstanding voltage (DWV):
Sleeving to case (min) - Not applicable.
Terminal to case (min) - 300 volts.

Insulation resistance:
Sleeving to case (min) - Not applicable.
Terminal to case (min) - 100 megohms.

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INCHES	MM
.005	.13
.025	.64
.031	.79
.138	3.51
.412	10.46
.437	11.10
.625	15.88
.812	20.62
.875	22.23
.969	24.61
1.062	26.97
1.312	33.32

Type designation	Case size	Dimensions			
		Dim	Min	Nom	Max
CL050SA13	A1	A	.846 (21.49)	.862 (21.89)	.878 (22.30)
	A2 1/	A	1.159 (29.44)	1.175 (29.84)	1.191 (30.25)
CL050SA33	A3	A	1.409 (35.79)	1.425 (36.20)	1.441 (36.60)
	A4 1/	A	1.784 (45.31)	1.800 (45.72)	1.816 (46.13)
	A5 1/	A	2.284 (58.01)	2.300 (58.42)	2.316 (58.82)

1/ CL050SD13, CL050SD23, and CL050SD33 brackets (of MIL-C-3965/9) may be used for case sizes A2, A4, and A5, respectively.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Spade lugs shall be rigid (either double welded, double riveted, single riveted and soldered, or single riveted and welded to the bracket).
5. Unless otherwise specified, tolerance is ±.062 (1.57 mm).

FIGURE 3. CL050 brackets.

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REQUIREMENTS:

Thermal shock:
DCL (max) - 125% of 25°C value of table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - 115% of initial requirement.

Solderability: Not applicable.

Terminal strength:
Pull test - In accordance with MIL-C-3965.
Bend test (if req.) - In accordance with MIL-C-3965.
Torque test - Not applicable.
Wire-lead bend test - Not applicable.
Twist test - Not applicable.

Surge voltage: See table I.

Moisture resistance:
DCL (max) - 125% of 25°C value of table I.
ΔCap - Within ±8% of initial measurement.
DF (max) - 115% of initial requirement.

Low temperature (storage):
DCL (max) - See table I (25°C).
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Stability at low and high temperatures:
Step 1 (+25°C):
DCL (max) - See table I.
Cap - Within tolerance of table I value.
DF (max) - Shall not exceed initial requirement.
Step 2 (-55°C):
Impedance - See table I.
ΔCap (max) - -35% of step 1 value.
Step 3 (+25°C):
DCL (max) - See table I.
ΔCap - Within ±5% of step 1 value.
DF (max) - Shall not exceed initial requirement.
Step 4 (+85°C):
DCL (max) - See table I.
ΔCap (max) - +25% of step 1 value.
DF (max) - Shall not exceed initial requirement.
Step 5 (+125°C):
DCL (max) - See table I.
ΔCap (max) - +25% of step 1 value.
DF (max) - Shall not exceed initial requirement.
Step 6 (+25°C):
DCL (max) - See table I.
ΔCap - Within ±5% of initial requirement.
DF (max) - Shall not exceed initial requirement.

Reverse voltage: In accordance with MIL-C-3965.

AC ripple life at 85°C: In accordance with MIL-C-3965.
Current - See table I.

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REQUIREMENTS:

Life at +85°C:
2,000-hr (qualification):
DCL (max) - See table I (25°C).
ΔCap - Within ±25% of initial measurement.
DF (max) - 130% of initial requirement.
240-hr (group B):
DCL (max) - See table I (25°C).
ΔCap - Within ±10% of initial measurement.
DF (max) - 130% of initial requirement.
1,760-hr (group C):
DCL (max) - See table I (25°C).
ΔCap - Within ±25% of initial measurement.
DF (max) - 130% of initial requirement.

Barometric pressure:
Test voltage - Rated voltage.

Life at +125°C:
DCL (max) - See table I (125°C and 25°C).
ΔCap - Within ±25% of initial measurement.
DF (max) - 130% of initial requirement.

Marking: Capacitors shall be marked as shown in the following example:
Example:
CL56BE961MPG - - Type designation.
12345 7615 - - - Source and date codes.
960 µF 15 Vdc - Capacitance and rated voltage (at 85°C).
Polarity - Plus (+) symbol near positive terminal (see figure 1).

Voltage group (85°C rated voltage:
I - - - - - - - - - 15 to 125V incl.

Custodian:
Air Force - 85

Review activity:
Air Force - 80

Preparing activity:
Air Force - 85

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MIL-C-3965/25 (USAF) Capacitors, Fixed, Electrolytic (Nonsolid Electrolyte) FSC 5910

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CONTRACT NUMBER

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